AP PRECALCULUS

Unit 1 • Lesson 1.6 practice • Answer key

Lesson 1.6 • Polynomial Functions and End Behavior

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

$\lim_{x\to\infty}\left(-4x^3+x^2\right)$ is

(A) ∞
(B) $-\infty$
(C) -4
(D) 0

Solution

Answer: (B). The leading term $-4x^3$ has odd degree and negative coefficient.

Question 2

Let $p(x)=(2-x)^3$. What is $\lim_{x\to-\infty} p(x)$?

(A) $-\infty$
(B) $+\infty$
(C) 8
(D) 0

Solution

Answer: (B). The leading term is $(-x)^3=-x^3$. As $x\to-\infty$, $-x^3\to+\infty$.

Question 3

Which polynomial increases without bound as $x\to\infty$ and as $x\to-\infty$?

(A) x^3+1
(B) $-x^4$
(C) $5x-x^2$
(D) $2x^6-x$

Solution

Answer: (D). Even degree with a positive leading coefficient.

Question 4

The graph of a polynomial falls to the left and rises to the right. Its degree and leading coefficient are

(A) even, positive
(B) even, negative
(C) odd, positive
(D) odd, negative

Solution

Answer: (C). Opposite end behaviors mean odd degree; rising to the right means positive leading coefficient.

Question 5 (Free response)

Let $f(x)=-2x^4+3x^3+x-7$ and $g(x)=x^2(3-x)(x+1)^2$.

(a) Write limit statements for the end behavior of f.
(b) Find the degree and leading term of g, and write limit statements for its end behavior.
(c) Could the graph of g have 5 turning points? Explain.

Solution
(a) Even degree, negative leading coefficient: $\lim_{x\to\infty} f(x) = -\infty$ and $\lim_{x\to-\infty} f(x) = -\infty$.
(b) Leading term $x^2 \cdot (-x) \cdot x^2 = -x^5$, degree 5. $\lim_{x\to\infty} g(x) = -\infty$ and $\lim_{x\to-\infty} g(x) = +\infty$.
(c) No. A polynomial of degree 5 has at most 4 turning points.